

A world of tilings

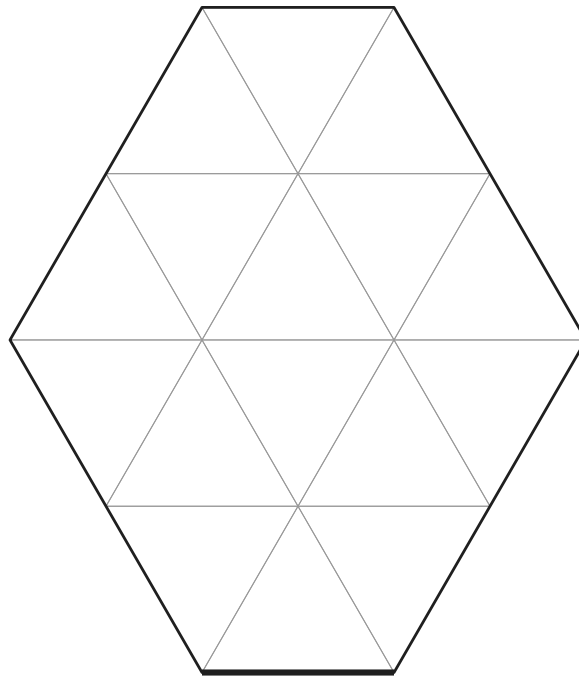
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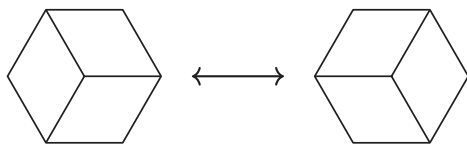
Use **eight small ($1\times$) blue rhombi**. Cover the outline exactly. Keep the bold bottom edge in the same place; different joins count as different tilings.

1. Build. Then change it.

Find a filling. Can you make a different filling with just one flip?



keep this edge at the bottom



The two fillings of a small hexagon.

One legal flip: Find three rhombi that fill a small regular hexagon. Lift only those three and refill that same hexagon in its other way.

Investigate: Can you get from any filling to any other by these tiny moves? How could you find out?

—
One small block edge

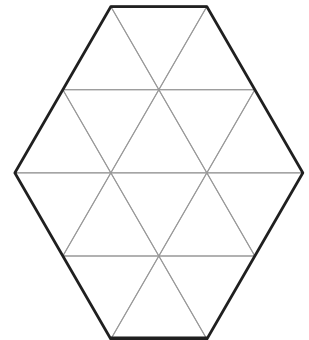
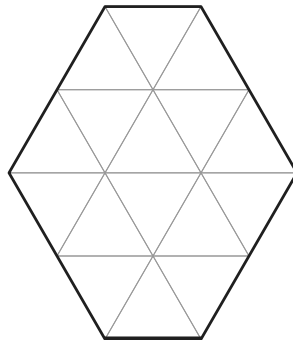
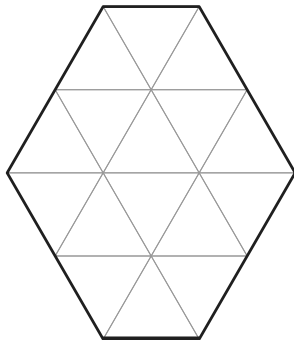
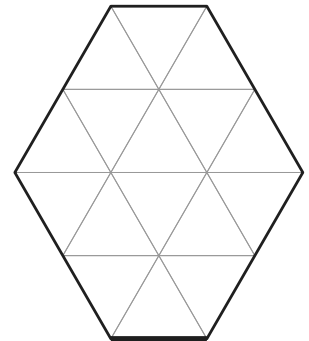
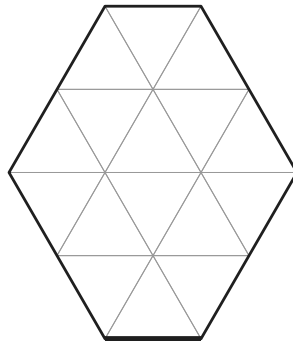
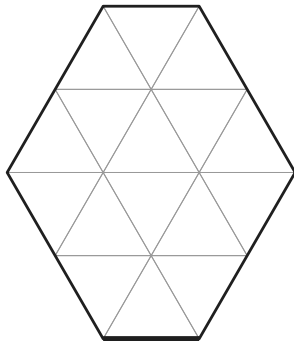
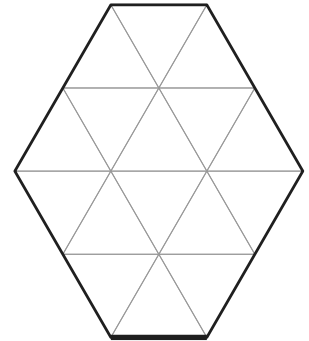
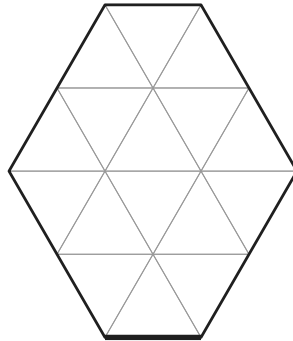
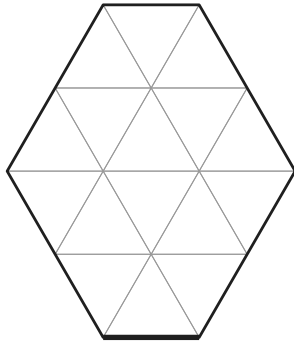
Adult: print **100% / Actual Size**, single-sided. Default edge: 1 inch (25.4 mm). Compare with a $1\times$ green triangle before using these mats.

How many tilings can you find?

Name: _____

Date: _____

Draw as many distinct tilings as you can, one on each blank board. Keep the bold edge at the bottom; different joins count as different tilings. These smaller boards are for drawing.



How many distinct tilings did you find? _____

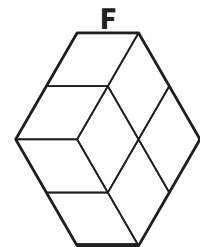
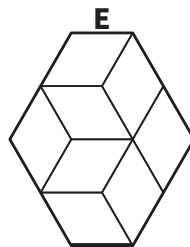
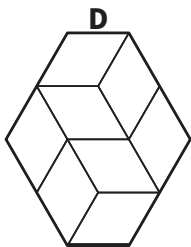
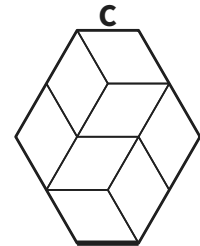
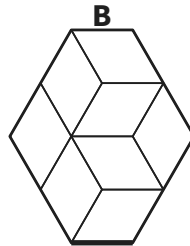
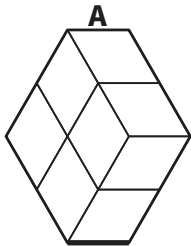
Build first. Explain by pointing, talking, or drawing.

Map the possible moves

Name: _____

Date: _____

Try finding tilings yourself first. Then use these six cards. The cards are a supplied list; explaining why no seventh tiling exists is a separate challenge.



2. Make a map.

Join two letters below **exactly when one legal flip** changes one tiling into the other. Use the blocks to check each line.

A

B

C

D

E

F

Can every card reach every other? Could a seventh tiling be hiding?

Build first. Explain by pointing, talking, or drawing.

Can you prove the best route?

Name: _____

Date: _____

3. From A to F.

Find the shortest route from A to F. Write the letters along your route.

My route: A

Why can no shorter route work?

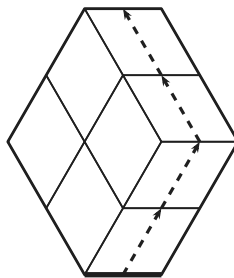
4. Odd or even?

Can you leave A and return to A in exactly three flips? Five flips? Any odd number? Give a reason that covers every route.

My reason:

A clue for going deeper: follow a ribbon.

Start at the bold bottom edge. Cross its rhombus to the opposite horizontal edge. Keep crossing upward until you reach the top. Each step goes left (L) or right (R).



This is A's ribbon. Record the ribbon for each card. How many orders of two L's and two R's are possible? What does a flip do to the order?

Build first. Explain by pointing, talking, or drawing.